

SYED MUHAMMAD AYYAN IBRAR

Lahore, Pakistan | +92 322 0621975 | syedmuhammadayyanibrar@gmail.com | [linkedin.com/in/ayyan-ibrar](https://www.linkedin.com/in/ayyan-ibrar) | github.com/syedmuhammadayyanibrar | syedmuhammadayyanibrar.github.io/resume

PROFESSIONAL SUMMARY

Enterprise AI Systems Engineer specializing in deterministic multi-agent federations, automated regulatory governance, and high-stakes decision workflows. Experienced in engineering autonomous contract lifecycle systems (CAS), EU AI Act compliance auditing (ComplianceOps), and multi-agent commercial deal bargaining using Python, FastAPI, LangGraph, Google Gemini, and PostgreSQL. Demonstrated expertise in eliminating hallucination risk in high-liability environments and accelerating commercial business cycles.

PROFESSIONAL EXPERIENCE

Freelance AI Systems Engineer | Self-Employed 2024 – Present

- Architected and deployed production-grade agentic AI pipelines and backend services for commercial clients, integrating LLMs with deterministic validation schemas and PostgreSQL persistence.
- Designed automated data-ingestion and validation workflows for heterogeneous data sources, reducing downstream processing overhead by 60% and enforcing cryptographic audit trails.
- Implemented dual-key human authorization safety controllers preventing unauthorized external API writes in enterprise integrations with Slack and Linear.

Advanced AI Bootcamp Fellow | Ghulam Ishaq Khan Institute (GIKI) • Grade A 2026

- Completed rigorous engineering program focused on applied multi-agent architectures, LLM orchestration, evaluation harnesses, and production MLOps deployment.

FEATURED ENTERPRISE PROJECTS

Contract Agentic Society (CAS) — Enterprise Contract Lifecycle Mesh github.com/.../CAS

Python 3.12, FastAPI, Google Gemini API, FastN MCP Nervous System, PostgreSQL, asyncpg, Docker, React

- Architected a federation of 6 autonomous agent societies (Contract, Risk, Negotiation, Compliance, Obligation, Dispute) to automate B2B contract lifecycles from intake to post-signature monitoring, **reducing review turnaround by 75%**.
- Engineered an adversarial Risk Intelligence debate pipeline (Prosecution vs. Defense) that stress-tests clauses, eliminating ungrounded alarms with **100% benchmark defense grounding**.
- Integrated enterprise nervous system via **FastN MCP workflows**, automating DocuSign intake, Slack risk escalation, and Google Calendar SLA milestone tracking.
- Built automated benchmark suite evaluating clause extraction completeness (100%) and liability calibration across diverse commercial agreements.

ComplianceOps — Evidence-Driven AI Compliance Auditor github.com/.../compliance-ops

LangGraph, Python, FastAPI, Next.js, pgvector, PostgreSQL, Slack API, Linear API

- Developed an autonomous compliance auditing platform for enterprise AI systems under the **EU AI Act**, eliminating manual multi-week audit cycles.
- Orchestrated stateful LangGraph workflows that gather evidence across GitHub repositories and Google Drive policies with **99.3% evidence accuracy**.
- Enforced deterministic safety gate requiring dual-key human approval before Linear remediation ticket execution, achieving **100.0% policy gate compliance** across 8 evaluation metrics.

Autonomous B2B Deal & Procurement Negotiation Engine github.com/.../negotiation_agent

LangGraph, Groq LLM, FastAPI, Redis Checkpoints, Qdrant Vector DB, PostgreSQL, LangSmith

- Engineered a decentralized multi-agent bargaining engine where buyer and vendor agents autonomously negotiate enterprise vendor contracts and cloud SLA agreements to prevent commercial deadlocks.
- Implemented an asynchronous **17-message state-machine protocol** with sequence enforcement, message deduplication, and coalition formation resolving deadlocked multi-party disputes.
- Developed behavioral trust modeling in Qdrant tracking counterparty concession velocity and contradictory claims across deal sessions, penalizing bad-faith posturing.

AuraSight — Edge-Native Voice Transaction & Accounting System github.com/.../AuraSight

Kotlin, Jetpack Compose, Whisper ONNX, Gemma 4, CameraX, SQLite / Room, Android SDK

- Built an offline-first multimodal retail transaction assistant enabling visually impaired shopkeepers to independently manage billing, currency verification, and credit ledgers (*Khata*) in Urdu.
- Designed constrained function-calling architecture enforcing deterministic schema execution for financial calculations, guaranteeing **zero arithmetic hallucinations**.
- Integrated local Whisper ONNX speech-to-text with physical hardware volume triggers for fast, reliable operation in noisy environments.

EDUCATION & HONORS

The Islamia University of Bahawalpur | Bachelor of Science in Artificial Intelligence • CGPA: 3.76 2022 – 2026

Honors: Recipient of Prime Minister's Laptop Scheme Award (National Merit Scholarship). Coursework: Multi-Agent Systems, Natural Language Processing, Deep Learning, Game Theory, Data Structures & Algorithms.

TECHNICAL SKILLS

Agent Orchestration:	Multi-Agent Federations, LangGraph, State Machines, Google Gemini API, FastN MCP, Tool-Calling, RAG, Adversarial Verification
Backend & Infrastructure:	Python 3.12 (Advanced AsyncIO), FastAPI, PostgreSQL, asyncpg, Redis, Qdrant, ChromaDB, pgvector, Docker, PyTest, CI/CD
Governance & Safety:	EU AI Act Compliance, Dual-Key HITL Authorization, Expected Calibration Error (ECE), Behavioral Trust Modeling, OWASP Top 10
Edge & Mobile AI:	On-Device Whisper ONNX, Gemma 4 Multimodal, CameraX, Kotlin, Jetpack Compose, SQLite / Room